
Part IV

Orthopedics of the Trunk

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12.1 Traumatic Spine Injuries

The formerly more frequent and dreaded ground fall has become less frequent, thanks to new belay devices, belaying trainings, and increasingly better protection of climbing areas. In boulder gyms, fractures of the spine are very rare due to the safety mats, and in climbing gyms, fall protection floors are increasingly used and belaying trainings are frequently given. This is shown by the accident statistics of alpine and climbing clubs [6–8]. Nevertheless, ground falls leading to serious injuries still occur occasionally. Falls into the climbing rope with wall collision seem to have increased as a consequence of the development of semiautomatic belaying devices. However, these falls usually result in injuries to the lower extremities. An exception is the “head over heels” fall, during which the climber turns and hits the wall with his skull or back. Particularly, severe hyperextension injuries of the spine can also result from a simple fall into the rope with high impact, if the climber falls unexpectedly, has had an unfavorable body position during the fall, or does not build up enough tension in the back during the fall.

12.1.1 Back Strain

Not only rope falls or sudden (rotational) movements but also a decrease in the strength of the muscles in a tense movement can lead to strains of the back, just as in other parts of the body. Mostly, these strains occur in eccentric movement. These injuries are small tears in muscles or ligaments that are rarely detectable in radiological imaging. Sometimes, sonography or MRI scans show increased fluid accumulation. Such strains can cause considerable pain when the damaged muscle or ligament is activated and, thus, impair performance for several weeks, although they are considered harmless. In this case, the main aim is to rule out the presence of more serious injuries. Therapeutically, relief can be achieved immediately after the accident by local cooling and in later phases through heat therapy and massages (e.g., through fascial rolls) as well as func-

tional physiotherapy. Ointments that promote blood circulation also have a positive effect. To prevent strains of the back, sufficient warming-up should be ensured before starting the high-performance part of the sport.

12.1.2 Fractures of the Spinous or Transverse Process

Isolated spinous and transverse process fractures are a rarity. There are reports of rarely seen clay shoveler’s fractures through sudden flexion force on the back and neck muscles [9]. Still, the majority of these fractures occur in direct impact trauma (ground fall or wall collision) or due to hyperflexion of the spine. More often, they occur in combination with other spinal column injuries. Therefore, if a spinal or transverse process fracture occurs in isolation, the presence of a second, previously overlooked spinal column injury must always be considered and searched for. In the case of frustrating conventional X-ray diagnostics, this can also be carried out with cross-sectional imaging. In the final analysis, MRI should be administered to exclude a ligamental injury.

The isolated spinal or transverse process fracture is a domain of conservative therapy. The leading symptom is pain, accompanied by swelling and, sometimes, hematoma. Immobilization is hardly possible because the pain is triggered by activation of the inserting muscles and their traction or by movement of the attached ribs. Therefore, sufficient pain therapy is absolutely necessary.

12.1.3 Hyperextension Fractures and Injuries

Hyperextension injuries are rare but highly severe injuries of the spine. They occur either in the thoracolumbar transition or in the cervical spine. Such injuries can occur when a climber falls into the rope at an unexpected moment, from an unfavorable body position, or when the body tension is low. In these cases, a tight pull of the rope by the

belayer results in a strong pull on the belay loop and, thus, a quasi-fixation of the pelvis and the lower lumbar spine. The upper body bends above the fixation, which leads to a frontal tearing and a rear compression of the spine. The injury can be purely ligamentous and transdiscal or completely or partially osseous. It is often accompanied by neurological deficits and usually associated with paraplegia. These unstable injuries require surgical treatment by ventral or dorsal stabilization. If there is also a sensorimotor deficit, this injury is an emergency, requiring immediate surgery.

As a purely ligamentous injury, hyperextension injuries can easily be overlooked in conventional X-rays. If the clinical symptoms and the mechanism of the accident show evidence for the suspicion of such an injury, cross-sectional imaging is always indicated.

Climbers with preexisting conditions such as ankylosing spondylitis or DISH (diffuse idiopathic skeletal hyperostosis) have an increased risk of suffering such an injury during a rope fall due to the reduced flexibility of the spine.

12.1.4 Compression and Burst Fractures

Compression and burst fractures occur as a result of climbing accidents with a ground fall or wall collision. Biomechanically, a distinction can be made between direct impact (floor or wall) and spinal compression or shearing. Torsion fractures, as they occur in high-speed traumas (e.g., traffic accidents), are an absolute exception. Clinically, patients with compression and burst fractures present with pain, swelling, hematoma, and, in severe cases, neurological deficits. Conventional X-ray diagnostics are usually followed by cross-sectional imaging (CT/MRI). For falls from a height of more than 3 m, cross-sectional imaging is always indicated (Fig. 12.1).

Conservative treatment can be administered for pure compression fractures without dislocation of the posterior vertebral body edge with a height reduction of less than 50% of the vertebral body and segmental kyphosis of less than 20%.

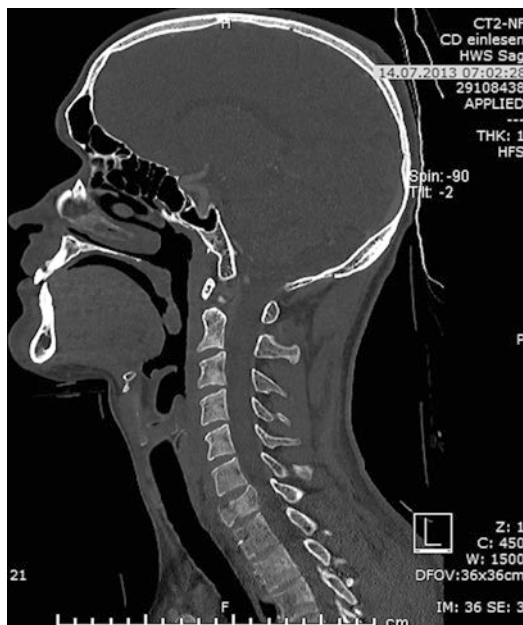


Fig. 12.1 Sagittal CT scan of a C7 burst fracture in a 30-year-old male climber before surgery

These are considered to be stable fractures. X-ray monitoring after 4–8 days and adequate pain therapy should be performed [10].

Compression fractures with posterior vertebral body edge dislocation, more than 50% height reduction of the vertebral body, or more than 20% segmental kyphosis are surgically addressed (Fig. 12.2). The same applies to burst fractures and all fractures with additional sensorimotor deficits. In the case of incomplete or complete paraplegia, this is a surgical emergency [10].

In the case of an isolated injury to a vertebral body, dorsal stabilization with a small-segment fixation is indicated. In this case, stabilization is performed in the vertebral body above and below the injured vertebra. If fractures occur in the thoracolumbar transition with distraction or torsion, two segments are used for stabilization (Figs. 12.3 and 12.4). If there is also a destruction of the intervertebral disc, ventral stabilization with the insertion of a bone graft or a vertebral body replacement is required.



Fig. 12.2 Radiographic image postoperative of the same patient after surgery with a ventral plate stabilization C6–Th1

Traumatic Spine Injuries

- **History:** Ground fall or wall collision, rarely rope fall or twisting motion
- **Onset:** Acute
- **Clinical:** Pain, swelling, hematoma, deformity, crepitations, sensorimotor deficit
- **Diagnostic:** Clinical finding, radiography, X-ray functional images, CT/MRI
- **Therapy:** In accordance with orthopedic trauma guidelines
- **Outcome:** Depending on the diagnosis, often bad with a primary sensorimotor deficiency

12.2 Chronic and Degenerative Back Diseases

The entire range of chronic and degenerative spinal diseases can be found among sport climbers as in the normal population—from easily treatable muscular tension, lumbalgia, and sciatica to conditions which are more difficult to treat, such as herniated discs and spondylolisthesis.



Fig. 12.3 Radiographic image of a Th12 compression fracture in a 26-year-old male climber due to ground fall, before surgery

Even though there are still too few comparative and reliable studies, it can be stated that sport climbing in general is a good training for the back [2, 4, 11]. The preventive value of sport climbing is shown in studies [12], as well as the fact that sport climbing has already been incorporated in rehabilitation and physiotherapy for a long time. Nevertheless, climbing without additional stretching and balancing exercises is a one-sided load that can manifest bad posture and associated suffering.

Special forms of chronic back problems in climbers are the “climber’s back” [3] and the “belayer’s neck” [5]. These are false postures developed by chronic overuse, uniform stress, and the biomechanical requirements of the climbing sport, which can lead to severe pain and degenerative changes in the spine.

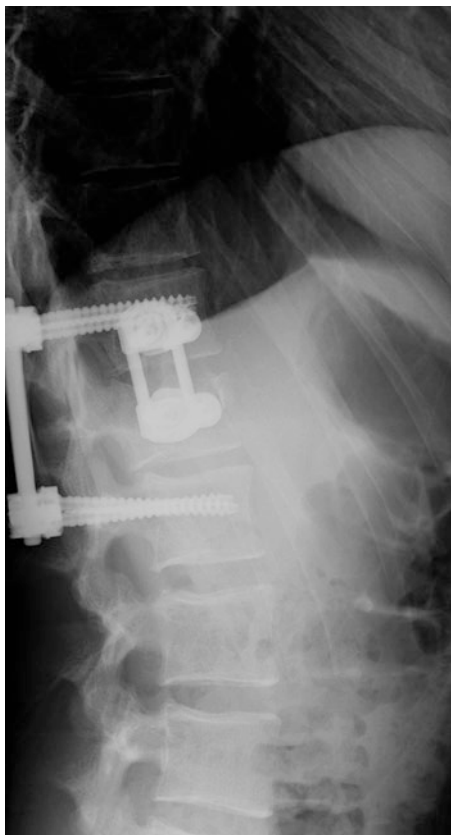


Fig. 12.4 Radiographic image of the same patient after surgery with short-segment posterior fixation Th11–L1 and accomplishment with anterior lumbar corpectomy via a mini-open, extreme lateral, transpsoas approach

It is not yet known how much damage is caused to the spine by the repetitive falls sport climbers take. It seems possible that the chronic microtrauma these falls create for the intervertebral discs could lead to early degeneration. This would be hard to prove, because most climbers who suffer from nucleus pulposus prolapse (a herniated disc) don't report it happening in a fall but when executing some sort of twisting motion.

12.2.1 Myalgia and Myogelosis

While myalgia simply describes the symptom of muscle pain which can occur in conjunction with many various diseases, myogelosis are palpable knots in muscles, which persist even when the

muscle returns to relaxation (e.g., in narcosis). This is the main difference to a simple muscular tension. The symptoms of myogelosis are palpable knots within the muscle, a tenderness to pressure, and muscle tightness. Histological findings showed atrophic muscle fibers and a thinning of the endomysium in these muscular knots, which is known to be a consequence of malfunctional metabolism through poor blood flow. Often through misuse or overuse of the back, the muscles become strained, a process which repeats itself in a vicious cycle. The muscles seize up from overuse and then, due to the stress of being flexed, receive poor blood flow. Lacking good nutrition, the muscle has to work a bit harder for any motion, increasing the tension and further decreasing the blood flow. This condition is especially common in the neck and shoulder area but also known for the lumbar spine. Until the muscles relax and receive blood flow and therefore a chance to recover, they will be stiff. A muscle that is too tight does not receive the appropriate amount of blood flow, which, in turn, allows the acids that are created during exercise to build up in the muscle body. Myogelosis can be painful and inhibit a climber's ability. If the condition is not treated as soon as possible, it can scar the muscle permanently. Through heating, massaging, and stretching the muscle, the excessive tightness can often be worked out and the blood flow can be stimulated. To prevent myogelosis from occurring during periods of intense training, it is important that climbers warm up correctly and then go through a "cooldown" period at the end of each workout [13].

12.2.2 Lumbago

The term lumbago subsumes a broad range of painful symptoms, leading to reduction in the mobility of the lumbar vertebrae, or lower spine. Most cases are harmless and easy to treat, but some rare back diseases can appear initially with lower back pain. Climbers experience lower back pain and reduction of mobility due to overuse and false postures or excessive training. This can also be the initial symptom of a "climber's back" [3].

For diagnosis, a clinical examination by an experienced physician might be enough, and radiography or cross-sectional imaging may not be required. Pathologies such as ankylosing spondylitis, Scheuermann's disease, disc herniations, kissing spine, spinal stenosis, scoliosis, and spondylolisthesis/spondylolysis must be ruled out when symptoms persist and present therapy-refractory. Therapy consists of mobility supporting physical therapy, heat therapy, and analgetic therapy.

12.2.3 Blocking of Facet Joints and Rip Joints

Many climbers report a blocking of joints, which frequently occurs and has a high tendency of reoccurrence. These blockings must be distinguished from joint dislocations [5]. To clarify this, a blocked joint is not actually a dislocated joint; this is only seen in trauma. Additionally, neither the small intervertebral joints nor the rib joints physically dislocate; the blockages are minor "dislocations" or shifts, which cannot be shown in cross-sectional imaging like CT or MRI scans. Increased stress to the chest spine, false postures, or sudden torsional movements can lead to stress onto the small intervertebral joints (facet joints). High stress to these facet joints leads to minor shifts, which stress their joint capsule and ligaments. Nevertheless, there is constant activation of these joints, leading to pain, discomfort, muscular tension, and sometimes even a decreased range of motion in parts of the spine. These symptoms might result in inflammation. The upper to middle thoracic vertebrae are affected the most. During climbing, the rhomboid muscles work hard to stabilize the scapula. If these muscles get weak and are no longer capable of stabilizing the scapula, it leads to the phenomenon of the "chicken winging" or "scapular winging" experienced by climbers during hard climbing sections. With the major and minor rhomboid muscle finding their insertions on the lower cervical and upper thoracic spine, this could also lead to the blocking of joints [5].

The increased kyphosis of a "climber's back" might predispose a climber to frequent blockages of the intervertebral joints, resulting in a decreased range of motion in certain parts of the spine. A proper clinical examination can differentiate between blockages, nerve compressions, and herniations. If a herniation or nerve compression has been excluded, therapy can be initiated. In early stages, a simple chiropractic maneuver can provide complete pain relief. Some patients report giving themselves the treatment, e.g., with two tennis balls or Blackroll® balls. If the blockage is already accompanied by muscular tension or inflammation, massages, heat therapy, and physiotherapy can help to ease the muscular tension. Often, spontaneous relief of the blockage occurs when the muscle tension is gone.

A similar pathology applies for the adjunct rib joints, meaning the proximal attachments of the ribs to the vertebra. A minor change of the positioning of the vertebrae together with the increased tension of the muscles leads to similar blockages of the rib joints. These blockings can be even more painful while breathing, as these joints move with every inhalation and exhalation. If these problems persist for a long time, they will lead to even more muscular tension and relieving postures.

12.2.4 Herniated Disc or Nucleus Pulposus Prolapsus

A herniated disc is more common in the lower back than in the mid back or the neck. The overall incidence is high. Intervertebral disc surgery is the most frequently performed spine surgery. Between each vertebra, there are shock absorbers that allow the spine to move but stay connected. These shock absorbers, the discs, are made up of a nucleus of thick liquid surrounded by an annular fibrous ring. A nucleus pulposus prolapsus occurs when pressure on the disc creates a bulge in the fibrous ring. This bulge can then put pressure on the nerve that branches out of the spine or on the spine itself. As previously mentioned, it is difficult to distinguish whether there has been a predisposition and a damage to the spine before



Fig. 12.5 Sagittal MRI T2 TSE scan of a 42-year-old male climber showing a L2–L3 disc herniation with a sequesterum

or if a certain activity (such as climbing) caused the prolapse. It is still not known how much damage is caused to the spine by the repetitive falls sport climbers take. It seems possible that the chronic microtrauma these falls cause to the intervertebral discs could lead to early degeneration. This would be hard to prove, because most climbers who suffer from nucleus pulposus prolapse (a herniated disc) don't report it happening in a fall but when doing some sort of twisting motion [14]. The classification differs between protrusion, prolapse, and a free sequestered disc. This results in typical symptoms causing pain, numbness, or weakness in according muscles. The key to a correct diagnosis is to relate the clinical findings (diminished reflexes, motor deficiency and lack/loss of sensibility) to the cross-sectional imaging findings (Fig. 12.5). If there is no congruence between clinical symptoms and MRI imaging, other pathologies (such as shingles neuralgia, multiple sclerosis, or peripheral nerve lesions) have to be considered. It is important to remember that many disc protrusions and even prolapses are completely asymptomatic. We have seen a lot of disc prolapses and protrusions as incidental findings on MRI scans in patients

who came complaining of other problems. Most cases (up to 90% of the lumbar disc herniations) respond well to conservative treatment. Analgetic therapy, physical therapy, and nerve root and epidural infiltration (steroids and local anesthetic) alleviate the patients' suffering. In severe cases with paralysis or persistent pain, surgery is inevitable.

12.2.5 Spondylolisthesis/ Spondylolysis

There is a rare condition known as spondylolisthesis, or “sliding vertebra,” which can be due to a birth defect, such as a dysplastic type, or mechanical overuse from athletics (e.g., in wrestlers, gymnasts, contortionists, football players, and climbers) [10]. Sports with causative mechanisms, such as lumbar hyperextension and rotation found in rowing, artistic gymnastics, and throwing sports, record high spondylolysis prevalence rates—16.88%, 16.96%, and 26.67%, respectively, compared to the general population's 3.0–8.02% [15]. Climbing also involves these causative mechanisms, and reports suggest that climbing may initiate the development of spondylolysis, but its prevalence has not yet been investigated [5]. Spondylolisthesis describes the condition of a ventral or dorsal shift of two vertebral bodies against each other, whereas spondylolysis describes the interruption of the pars interarticularis of the vertebral body. There is a genetic predisposition to dysplastic type of spondylolisthesis, which can also occur at the end of multi-segmental vertebral body fusions (after surgery) or as a consequence of Scheuermann's disease. Patients suffer from lumbar pain during movement, especially reclination; there may also be radiculopathy. Diagnosis is made by functional X-ray in extension and flexion to detect segmental hypermobility (Fig. 12.6). Cross-sectional imaging like CT and MRI can display the interruption of the pars interarticularis of the vertebral body (Fig. 12.7). In severe cases, surgery as a small-segment instrumentation is necessary. For climbers diagnosed with spondylolisthesis graded Meyerding I and II but not suffering from



Fig. 12.6 Radiographic image of a young male climber (9 years old) with a congenital spondylolysis/spondylolysis L5/S1 graded Meyerding II



Fig. 12.7 MRI STIR sequence scans of the same boy show the spondylolysis in the pars interarticularis L5/S1 graded Meyerding I

any symptoms, we recommend X-ray controls in an annual interval. For these climbers, it is not necessary to stop climbing.

12.2.6 Thoracic Outlet Syndrome

Excessive tension of a shortened pectoralis minor muscle or scalene muscle can lead to an entrapment of the nerves and blood vessel which run into the arm. Though rare, thoracic outlet syndrome is a nerve or blood vessel compression syndrome that affects the shoulder and arm. Overdevelopment and tightening of the small chest muscles and the scalene muscles (in the neck) can put pressure on the brachial plexus or/and subclavian artery [16]. Swelling of muscles after stress, overstretching, or anatomic abnor-

malities might be the reasons for this syndrome; for example, people with an extra rib in their upper body can also suffer from this syndrome. If the artery is compressed, paleness and cold skin are symptoms due to reduced blood flow. Compression of the nerve might lead to diffused pain in the arms, numbness, or a heavy sensation of the hands and lower arms. The condition is most noticeable when the arms are raised above the head. The symptoms vary according to whether the muscles are placing pressure on the nerves or the blood vessels. The syndrome can be diagnosed with the Adson test in a clinical examination or provocation duplex sonography.

Intensive physical therapy such as stretching the pectoralis and scalene muscles and the thoracic spine can often reduce the pain. Surgery is only necessary in very severe cases.

Thoracic Outlet Syndrome

- **History:** Congenital (additional rib), overuse, muscular disbalance
- **Onset:** Chronic
- **Clinical:** Pain in the arms, numbness
- **Diagnostic:** Clinical findings, Adson test, provocation duplex sonography
- **Therapy:** Conservative—stretching and balancing exercises, surgery in severe cases only
- **Outcome:** Good

12.2.7 Climber's Back

“Climber’s back” is characterized by anterior positioned shoulders and a “humpback” spine. The presentation of this postural adaptation is a combination of hyperkyphosis of the thoracic spine and a hyperlordosis of the lumbar spine, as well as a recurvated body hiatus [5, 17], which is similar to “swimmer’s back” [18, 19]. The origin of these spinal postural dysfunctions is probably a functional adaptation of the form of the spine designed to meet the sport-specific biomechanical stresses of the sport [20, 21]. It is thought that shortened pectoralis muscles are one of the main factors leading to this spinal postural dysfunction [17, 22]. Our studies showed [3] that the climbing ability level was strongly correlated to the postural adaptations found frequently in elite male climbers who climb routes graded UIAA 10 or above. Given the greater differences of both the kyphosis and lordosis angles between a sport climbers group and a recreational climbers group (control group) in our study, it is suggested that an even greater difference between performance-oriented climbers and a normal population is likely. We only examined male climbers in our study. In female climbers, there seem to be different postural adaptations to climbing; this might be due to their different muscle morphology as well as their different climbing technique [22].

Although sport climbing has been used in physical therapy for back disorders and rehabilitation as well as physiotherapy for scoliosis [23,

24], excessive training and/or the neglect of compensatory exercises may lead to the development of “climber’s back.” The functional adaptation of the thoracolumbar spine to specific sporting exposure is not a performance precursory variance of the norm but has a latent disease potential, especially during the growth period [19, 25, 26]. The biomechanical adaptations to intensive climbing that can result in a “climber’s back” may cause an aberrant erect posture but may be an essential adaptation to facilitate elite climbing kinetics. Nevertheless, we think that this adaptation to the specific postural requirements of higher-grade climbing is not advantageous or a necessary prerequisite of elite climbing kinetics; we feel that it is a precursor to the associated health conditions brought on by the development of thoracic outlet/inlet syndrome or early joint disorders due to the increased stress onto the pars interarticularis of the spine [21]. Therefore, it is not only to be seen as a physiological adaptation of the specific postural requirements of intensive climbing but also as a pathological finding, which demands compensatory training and exercise, especially in adolescent athletes (Fig. 12.8). A difference in the length of the legs (a trait most people unknowingly have) and congenital problems (like scoliosis) can contribute to false posture. Climbers developing thoracic hyperkyphosis, lumbar hyperlordosis, and anterior positioned shoulders as well as shortened pectoralis muscles might experience back or shoulder pain, reduced mobility, and sometimes even radiculopathy. In clinical examination, an appearance of the characteristic posture with thoracic hyperkyphosis and lumbar hyperlordosis should lead to examination of shortening of pectoralis muscles (e.g., using protocol given by Janda). Preventative stretching exercises (for pectoralis and iliopsoas muscles) should be recommended to the athletes. Here, we refer to our ACT (Adjunct Compensatory Training for Rock Climbers, www.act.clinic), which provides exercises for stretching and balancing the spine and core musculature.



Fig. 12.8 Typical climber posture (climber's back): overextension of the vertebral spine and a concave chest. An elite climber displaying the hyperkyphosis and hyperlordosis characteristic of a "climber's back" (reprinted with permission)

12.2.8 "Belayer's Neck" and Cervical Spine Syndrome

Cervical spine syndrome or cervical pain syndrome is a well-known and widespread set of symptoms. The "belayer's neck" is a postural adaption to the constant overextension the cervical spine experiences in hours of belaying and climbing. This could be worsened by the increased thoracic kyphosis from a "climber's back," leading to a compensatory overextension of the cervical spine. In combination with shortened pectoralis muscles, this brings high stress onto the cervical spine. This can become a problem while belaying long and steep routes and spending a lot of time belaying and climbing. Due to the increased overextension of the cervical spine, high pressure is applied to the small intervertebral joints. This irritates the joint cartilage and joint capsule. As a

protective mechanism, the paravertebral muscles raise their tension to protect the spine, resulting in cervical spine syndrome-like symptoms. Even though many climbers and belayers suffer from these symptoms, there is almost no scientific work on the consequences and circumstances of "belayer's neck." The range of symptoms includes neck and back pain, headaches, or even paresthesia and hypesthesia (prickling sensation and numbness), as well as impaired vision and ringing in the ears. A proper clinical examination (e.g., Spurling test) by a physician can exclude more serious pathologies, such as herniated discs or nerve root compression and can lead to diagnosis. In doubt, MRI can provide diagnosis, whereas conventional radiography is obsolete. The therapeutic regimen covers analgetic, massage, and physical therapy as well as local infiltration and muscle relaxant medication.

For prevention purposes, prism glasses are available for purchase, e.g., CU belay glasses®, LACD belay glasses®, or Y&Y glasses®, which can be highly recommended for people with cervical spine problems (Fig. 12.9). Through mirroring prisms, the belayer can have an image of the climber without keeping his head in an upright and overextended position (Fig. 12.10). Using prism glasses requires initial training and requires some time for adjustment, which is why we recommend starting to belay with prism glasses in top rope climbing.

Climber's Back and Belayer's Neck

- **History:** Overuse, bad postures, and functional adaptation to sport-specific stress
- **Onset:** Chronic
- **Clinical:** Lumbar or cervical spine pain, typical humpback, shortened pectoralis, and iliopsoas muscle
- **Diagnostic:** Clinical findings, typical humpback with hyperkyphosis of the thoracic and a hyperlordosis of the lumbar spine
- **Therapy:** Conservative—stretching and balancing exercises, prism glasses (belayer's neck)
- **Outcome:** Good, although can become chronic with degenerative injuries



Fig. 12.9 Female climber belaying with CU belay glasses®. (Reprinted with permission)



Fig. 12.10 Prism glasses CU belay glasses® (reprinted with permission)

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